

Randomized Algorithms (2026 Spring)

Final Exam: Detailed Solutions

Problem 1. Revised-Rand-Median

This revised algorithm has been shown in the lecture. Basically it's still a linear-time algorithm, yet has asymptotically lower failure probability.

Let

$$s = n^{5/6}, \quad t = n^{2/3}.$$

The algorithm samples s elements, sorts them, and defines

$$d = \left(\frac{s}{2} - t\right)\text{-th smallest sampled element}, \quad u = \left(\frac{s}{2} + t\right)\text{-th smallest sampled element}.$$

For clarity, we ignore floor and ceiling effects. They only change constants and do not affect the asymptotic bounds.

Time complexity (10%)

Sampling R takes $O(s) = O(n^{5/6})$ time. Sorting R takes

$$O(s \log s) = O(n^{5/6} \log n) \text{ time.}$$

Comparing every element of S with d and u takes $O(n)$ time. Finally, if the algorithm does not fail because of a large candidate set, then $|C| \leq 4n^{5/6}$, so sorting C takes

$$O(|C| \log |C|) \leq O(n^{5/6} \log n).$$

Therefore the total running time is $O(n + n^{5/6} \log n) = O(n)$. Thus the algorithm runs in linear time.

Failure probability: Chebyshev-style proof (15%)

Assume, for simplicity, that all elements are distinct and that n is even. Let the median have rank $n/2$. The algorithm may output FAIL because

- (i) d is above the median, so $l_d > n/2$;
- (ii) u is below the median, so $l_u < n/2$;
- (iii) the candidate set $C = \{x \in S : d \leq x \leq u\}$ has size larger than $4n^{5/6}$.

Event 1: d is above the median. Let X be the number of sampled elements from the lower half of S . Then

$$X \sim \text{Bin}(s, 1/2), \quad \mathbb{E}[X] = s/2, \quad \text{Var}(X) = s/4.$$

If d is above the median, then fewer than $s/2 - t$ sampled elements lie in the lower half. Hence

$$X < \frac{s}{2} - t.$$

By Chebyshev's inequality,

$$\Pr \left[X < \frac{s}{2} - t \right] \leq \Pr [|X - \mathbb{E}[X]| \geq t] \leq \frac{\text{Var}(X)}{t^2} = \frac{s/4}{t^2}.$$

Since $s = n^{5/6}$ and $t = n^{2/3}$,

$$\frac{s/4}{t^2} = \frac{n^{5/6}}{4n^{4/3}} = \frac{1}{4\sqrt{n}}.$$

Therefore

$$\Pr[l_d > n/2] \leq \frac{1}{4\sqrt{n}}.$$

Event 2: u is below the median. By the symmetric argument, if u is below the median, then too many sampled elements lie in the lower half. Again Chebyshev's inequality gives

$$\Pr[l_u < n/2] \leq \frac{1}{4\sqrt{n}}.$$

Event 3: $|C| > 4n^{5/6}$. Let a be the element of rank

$$\frac{n}{2} - 2n^{5/6},$$

and let b be the element of rank

$$\frac{n}{2} + 2n^{5/6}.$$

If $d \geq a$ and $u \leq b$, then $C \subseteq [a, b]$, and therefore

$$|C| \leq 4n^{5/6}$$

up to harmless rounding. Hence

$$\{|C| > 4n^{5/6}\} \subseteq \{d < a\} \cup \{u > b\}.$$

To bound $\Pr[d < a]$, let Y be the number of sampled elements with rank at most $n/2 - 2n^{5/6}$. Then

$$Y \sim \text{Bin} \left(s, \frac{1}{2} - 2n^{-1/6} \right),$$

so

$$\mathbb{E}[Y] = s \left(\frac{1}{2} - 2n^{-1/6} \right) = \frac{s}{2} - 2n^{2/3} = \frac{s}{2} - 2t.$$

The event $d < a$ implies

$$Y \geq \frac{s}{2} - t = \mathbb{E}[Y] + t.$$

Using Chebyshev's inequality,

$$\Pr[d < a] \leq \Pr[|Y - \mathbb{E}[Y]| \geq t] \leq \frac{\text{Var}(Y)}{t^2} \leq \frac{s/4}{t^2} = \frac{n^{5/6}}{4n^{4/3}} = \frac{1}{4\sqrt{n}}.$$

Similarly,

$$\Pr[u > b] \leq \frac{1}{4\sqrt{n}}.$$

Therefore

$$\Pr[|C| > 4n^{5/6}] \leq \frac{1}{2\sqrt{n}}.$$

Combining the three failure types by the union bound,

$$\Pr[\text{FAIL}] \leq \frac{1}{4\sqrt{n}} + \frac{1}{4\sqrt{n}} + \frac{1}{2\sqrt{n}} = \frac{1}{\sqrt{n}}.$$

Thus a standard lecture-style Chebyshev bound is

$$\boxed{\Pr[\text{FAIL}] \leq n^{-1/2}.}$$

A sharper Chernoff/Hoeffding bound

Claim 1. *Using Hoeffding's inequality instead of Chebyshev, for a binomial variable with s trials,*

$$\Pr[|X - \mathbb{E}[X]| \geq t] \leq \exp\left(-\frac{2t^2}{s}\right).$$

Proof. Let

$$X \sim \text{Bin}(s, p).$$

Equivalently, we may write

$$X = X_1 + X_2 + \cdots + X_s,$$

where $X_1, X_2, \dots, X_s$ are independent Bernoulli random variables with

$$\Pr[X_i = 1] = p, \quad \Pr[X_i = 0] = 1 - p.$$

Thus,

$$\mathbb{E}[X] = sp.$$

Hoeffding's inequality states that if $X_1, X_2, \dots, X_s$ are independent random variables satisfying

$$X_i \in [a_i, b_i] \quad (1 \leq i \leq s),$$

then for every $t > 0$,

$$\Pr[X - \mathbb{E}[X] \geq t] \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^s (b_i - a_i)^2}\right),$$

where $X = \sum_{i=1}^s X_i$. Since each Bernoulli variable satisfies $X_i \in [0, 1]$, we have $b_i - a_i = 1$. Therefore,

$$\sum_{i=1}^s (b_i - a_i)^2 = \sum_{i=1}^s 1^2 = s.$$

Substituting into Hoeffding's inequality yields

$$\Pr[X - \mathbb{E}[X] \geq t] \leq \exp\left(-\frac{2t^2}{s}\right).$$

Similarly,

$$\Pr[X - \mathbb{E}[X] \leq -t] \leq \exp\left(-\frac{2t^2}{s}\right).$$

□

OK, now we apply Claim 1 to the problem. Since

$$\frac{t^2}{s} = \frac{n^{4/3}}{n^{5/6}} = n^{1/2},$$

each of the four bad deviations used above has probability at most $e^{-2\sqrt{n}}$. Therefore a sharper upper bound is

$$\boxed{\Pr[\text{FAIL}] \leq 4e^{-2\sqrt{n}}.}$$

This is exponentially smaller than the Chebyshev bound. Just for your reference. Getting the Chebyshev's bound is already good.

Problem 2. Alice wins fewer than $n/2$ games

Let

X = the number of games Alice wins.

Then

$$X \sim \text{Bin}(n, 0.6), \quad \mu = \mathbb{E}[X] = 0.6n = \frac{3n}{5}.$$

We want to bound

$$\Pr[X < n/2].$$

Write

$$\frac{n}{2} = (1 - \delta)\mu.$$

Since $\mu = 3n/5$,

$$(1 - \delta)\frac{3n}{5} = \frac{n}{2},$$

and hence

$$1 - \delta = \frac{5}{6}, \quad \delta = \frac{1}{6}.$$

By the multiplicative Chernoff lower-tail bound,

$$\Pr[X < (1 - \delta)\mu] \leq \exp\left(-\frac{\delta^2\mu}{2}\right).$$

Therefore

$$\Pr[X < n/2] \leq \exp\left(-\frac{(1/6)^2(0.6n)}{2}\right) = \exp\left(-\frac{n}{120}\right).$$

Thus

$$\boxed{\Pr[X < n/2] \leq e^{-n/120}.}$$

Problem 3. Isolated vertices in $G_{n,N}$

Let

$$N = \left\lfloor \frac{n \ln n + cn}{2} \right\rfloor, \quad c > 0.$$

Let X be the number of isolated vertices. Write

$$X = \sum_{v=1}^n I_v,$$

where $I_v = 1$ if vertex v is isolated and $I_v = 0$ otherwise. By linearity of expectation,

$$\mathbb{E}[X] = n \Pr[v \text{ is isolated}].$$

The total number of possible edges is

$$M = \binom{n}{2}.$$

For a fixed vertex v , there are $n - 1$ edges incident to v . To make v isolated, all N chosen edges must come from the remaining

$$M - (n - 1) = \binom{n - 1}{2}$$

edges. Hence

$$\Pr[v \text{ is isolated}] = \frac{\binom{M - (n - 1)}{N}}{\binom{M}{N}}.$$

Therefore

$$\mathbb{E}[X] = n \frac{\binom{M - (n - 1)}{N}}{\binom{M}{N}}.$$

Now

$$\frac{\binom{M - (n - 1)}{N}}{\binom{M}{N}} = \prod_{i=0}^{N-1} \left(1 - \frac{n - 1}{M - i}\right).$$

Taking logarithms,

$$\ln \Pr[v \text{ is isolated}] = \sum_{i=0}^{N-1} \ln \left(1 - \frac{n - 1}{M - i}\right).$$

Since $N = O(n \ln n)$ and $M = \Theta(n^2)$, we have $i = o(M)$ throughout the sum. Using $\ln(1 - x) = -x + O(x^2)$,

$$\ln \Pr[v \text{ is isolated}] = -(n - 1) \sum_{i=0}^{N-1} \frac{1}{M - i} + O \left(\sum_{i=0}^{N-1} \frac{(n - 1)^2}{(M - i)^2} \right).$$

The error term is

$$O \left(N \cdot \frac{n^2}{n^4} \right) = O \left(\frac{\ln n}{n} \right) = o(1).$$

Also,

$$(n - 1) \sum_{i=0}^{N-1} \frac{1}{M - i} = (n - 1) \frac{N}{M} + o(1).$$

Since $M = n(n - 1)/2$ and

$$N = \frac{n \ln n + cn}{2} + O(1),$$

we get

$$(n - 1) \frac{N}{M} = (n - 1) \frac{(n \ln n + cn)/2 + O(1)}{n(n - 1)/2} = \ln n + c + o(1).$$

Therefore

$$\ln \Pr[v \text{ is isolated}] = -\ln n - c + o(1),$$

and so

$$\Pr[v \text{ is isolated}] = \frac{e^{-c+o(1)}}{n}.$$

Finally,

$$\mathbb{E}[X] = n \cdot \frac{e^{-c+o(1)}}{n} = e^{-c+o(1)}.$$

Thus

$$\boxed{\mathbb{E}[X] \rightarrow e^{-c}.$$

Problem 4. Coupon collector failure probability after $R = 2cn \ln n$ draws

There are n coupon types. Let A_i be the event that coupon type i is not collected after R draws. For a fixed type i ,

$$\Pr[A_i] = \left(1 - \frac{1}{n}\right)^R.$$

Using $1 - x \leq e^{-x}$,

$$\Pr[A_i] \leq e^{-R/n}.$$

Since $R = 2cn \ln n$,

$$e^{-R/n} = e^{-2c \ln n} = n^{-2c}.$$

The failure event is $A_1 \cup \dots \cup A_n$. By the union bound,

$$\Pr[\text{failure}] \leq \sum_{i=1}^n \Pr[A_i] \leq n \cdot n^{-2c} = n^{1-2c}.$$

Therefore

$$\boxed{\Pr[\text{failure}] \leq n^{1-2c}.$$

Since $c > 1/2$, this bound tends to 0, which is asymptotically smaller than the Markov bound $1/2$.

Problem 5. Bad pivot probability in QuickSort

Assume n is divisible by 3. The bad event is that the pivot rank is

$$< \frac{n}{3} \quad \text{or} \quad > \frac{2n}{3}.$$

Part (a). One uniformly random pivot

Ranks smaller than $n/3$ are

$$1, 2, \dots, \frac{n}{3} - 1,$$

so there are $n/3 - 1$ such ranks. Ranks larger than $2n/3$ are

$$\frac{2n}{3} + 1, \dots, n,$$

so there are $n/3$ such ranks. Therefore

$$\Pr[\text{bad pivot}] = \frac{(n/3 - 1) + (n/3)}{n} = \frac{2}{3} - \frac{1}{n}.$$

Thus

$$\boxed{\Pr[\text{bad pivot}] \leq \frac{2}{3}.$$

Part (b). Median of three independently chosen elements

Let the three sampled ranks be independent and uniform from $\{1, \dots, n\}$. The median is smaller than $n/3$ only if at least two of the three samples have rank smaller than $n/3$.

Let

$$q_L = \Pr[\text{one sample has rank} < n/3] = \frac{n/3 - 1}{n} = \frac{1}{3} - \frac{1}{n}.$$

Then

$$\Pr[\text{median} < n/3] = \binom{3}{2} q_L^2 (1 - q_L) + q_L^3 = 3q_L^2 - 2q_L^3.$$

Since $q_L < 1/3$,

$$\Pr[\text{median} < n/3] < 3 \left(\frac{1}{3}\right)^2 - 2 \left(\frac{1}{3}\right)^3 = \frac{7}{27}.$$

Similarly,

$$q_H = \Pr[\text{one sample has rank} > 2n/3] = \frac{n/3}{n} = \frac{1}{3}.$$

Thus

$$\Pr[\text{median} > 2n/3] = \binom{3}{2} q_H^2 (1 - q_H) + q_H^3 = 3 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right) + \left(\frac{1}{3}\right)^3 = \frac{7}{27}.$$

The two bad events are disjoint, so

$$\Pr[\text{bad pivot}] < \frac{7}{27} + \frac{7}{27} = \frac{14}{27}.$$

Therefore

$$\Pr[\text{bad pivot}] \leq \frac{14}{27}.$$

This is better than the one-sample bound $2/3$.

Problem 6. Simulating a fair coin from a biased coin

Use the von Neumann trick.

Algorithm.

1. Toss the biased coin twice.
2. If the result is HT , output HEAD.
3. If the result is TH , output TAIL.
4. If the result is HH or TT , discard both tosses and repeat.

Let the biased coin show Head with probability p , where $0 < p < 1$. Then

$$\Pr[HT] = p(1 - p),$$

and

$$\Pr[TH] = (1 - p)p.$$

Therefore

$$\Pr[HT] = \Pr[TH].$$

The events HT and TH are the only events on which the algorithm stops in a given round. Conditional on stopping,

$$\Pr[\text{output Head}] = \frac{\Pr[HT]}{\Pr[HT] + \Pr[TH]} = \frac{p(1-p)}{p(1-p) + (1-p)p} = \frac{1}{2}.$$

Similarly,

$$\Pr[\text{output Tail}] = \frac{1}{2}.$$

The probability of stopping in each round is

$$\Pr[HT \text{ or } TH] = 2p(1-p) > 0.$$

Thus the algorithm stops with probability 1, and when it stops, it outputs Head with probability $1/2$. Therefore the algorithm correctly simulates a fair coin.

Problem 7. Lovász Local Lemma application to k -SAT

Assume each clause contains exactly k distinct variables, and each variable appears in at most

$$T = \frac{2^k}{4k}$$

clauses, counting both positive and negative occurrences.

Choose a truth assignment uniformly at random. For each clause C , define the bad event

$$A_C = \{C \text{ is not satisfied}\}.$$

Since C contains k distinct variables, exactly one of the 2^k assignments to those variables falsifies the clause. Therefore

$$\Pr[A_C] = 2^{-k}.$$

Thus

$$p = 2^{-k}.$$

Next, we bound the dependency degree. A bad event A_C is mutually independent of all bad events corresponding to clauses sharing no variables with C . Clause C contains k variables. Each variable appears in at most T clauses total. Hence the number of other clauses that can share a variable with C is at most

$$d \leq kT = k \cdot \frac{2^k}{4k} = \frac{2^k}{4}.$$

Using the symmetric Lovász Local Lemma in the form

$$4pd \leq 1,$$

we obtain

$$4pd \leq 4 \cdot 2^{-k} \cdot \frac{2^k}{4} = 1.$$

Therefore the Local Lemma implies

$$\Pr \left[\bigcap_C \overline{A_C} \right] > 0.$$

That is, with positive probability, no clause is violated. Hence there exists a truth assignment satisfying all clauses. Therefore the formula is satisfiable.

If every variable appears in at most $\left\lceil \frac{2^k}{4k} \right\rceil$ clauses, then the k -SAT formula is satisfiable.

Problem 8. Randomized local search for satisfiable 2-SAT

Let S be a fixed satisfying assignment. Let

$$X_i = \#\{x_\ell : A_i(x_\ell) = S(x_\ell)\}.$$

Thus X_i counts how many variables currently agree with the satisfying assignment S . We are given that whenever $0 < X_i < n$, one repair step increases X_i by 1 with probability at least $1/2$.

Part (a). System of equations for the pessimistic Markov chain

The pessimistic Markov chain (Y_i) has states $0, 1, \dots, n$. State n is absorbing for the hitting-time calculation. Let

$$h_j = \mathbb{E}[\text{number of steps to reach } n \mid Y_0 = j].$$

Clearly,

$$h_n = 0.$$

From state 0, the chain moves to 1 with probability 1. Hence

$$h_0 = 1 + h_1.$$

For $1 \leq j \leq n-1$, the chain moves to $j+1$ with probability $1/2$ and to $j-1$ with probability $1/2$. Hence

$$h_j = 1 + \frac{1}{2}h_{j+1} + \frac{1}{2}h_{j-1}.$$

Therefore the system is

$$\begin{cases} h_n = 0, \\ h_0 = 1 + h_1, \\ h_j = 1 + \frac{1}{2}h_{j+1} + \frac{1}{2}h_{j-1}, \quad 1 \leq j \leq n-1. \end{cases}$$

Part (b). Solving the system

For $1 \leq j \leq n-1$,

$$h_j = 1 + \frac{1}{2}h_{j+1} + \frac{1}{2}h_{j-1}.$$

Rearranging,

$$h_{j+1} - 2h_j + h_{j-1} = -2.$$

This suggests that h_j is a quadratic function of j . Let

$$h_j = -j^2 + Aj + B.$$

Using the boundary condition $h_n = 0$,

$$0 = -n^2 + An + B.$$

Using $h_0 = 1 + h_1$, we have

$$B = 1 + (-1 + A + B) = A + B.$$

Thus $A = 0$. Then $h_n = 0$ gives

$$0 = -n^2 + B,$$

so $B = n^2$. Therefore

$$h_j = n^2 - j^2.$$

In particular,

$$h_0 = n^2.$$

Since $h_j \leq n^2$ for every j , the pessimistic chain reaches n in expected time at most n^2 .

Now compare this with the randomized 2-SAT local search. Whenever $0 < X_i < n$, the probability that X_i increases is at least $1/2$, whereas the pessimistic chain increases with probability exactly $1/2$. Thus the actual process is no slower than the pessimistic chain. Therefore the expected number of repair steps is at most n^2 .

Problem 9. Birthday problem

There are m people and n possible birthdays, with $m \leq n$. Each birthday is chosen independently and uniformly.

The first person may have any birthday. For the second person to avoid a collision, they must avoid the first person's birthday, so the conditional probability is

$$1 - \frac{1}{n}.$$

For the third person to avoid a collision, they must avoid the birthdays of the first two people, so the conditional probability is

$$1 - \frac{2}{n}.$$

Continuing in this way, the $(j + 1)$ -st person must avoid j already-used birthdays, so the conditional probability is

$$1 - \frac{j}{n}.$$

Therefore the probability of no shared birthday is

$$\prod_{j=1}^{m-1} \left(1 - \frac{j}{n}\right).$$

Equivalently,

$$\Pr[\text{no collision}] = \frac{n(n-1)(n-2)\cdots(n-m+1)}{n^m}.$$

Approximation of the product. From the first part, the probability that no two people share the same birthday is

$$\Pr[\text{no collision}] = \prod_{j=1}^{m-1} \left(1 - \frac{j}{n}\right).$$

To approximate this product, we use the standard approximation

$$1 - x \approx e^{-x} \quad \text{when } x \text{ is small.}$$

Therefore,

$$1 - \frac{j}{n} \approx e^{-j/n}.$$

Substituting this approximation into the product gives

$$\prod_{j=1}^{m-1} \left(1 - \frac{j}{n}\right) \approx \prod_{j=1}^{m-1} e^{-j/n}.$$

Since a product of exponentials is the exponential of the sum of the exponents, we have

$$\prod_{j=1}^{m-1} e^{-j/n} = \exp \left(- \sum_{j=1}^{m-1} \frac{j}{n} \right).$$

Now,

$$\sum_{j=1}^{m-1} j = \frac{m(m-1)}{2}.$$

Thus,

$$\exp \left(- \sum_{j=1}^{m-1} \frac{j}{n} \right) = \exp \left(- \frac{m(m-1)}{2n} \right).$$

When m is large, we have $m(m-1) \approx m^2$. Hence,

$$\Pr[\text{no collision}] = \prod_{j=1}^{m-1} \left(1 - \frac{j}{n} \right) \approx \exp \left(- \frac{m(m-1)}{2n} \right) \approx e^{-m^2/(2n)}.$$

Therefore,

$\Pr[\text{no two people share the same birthday}] \approx e^{-m^2/(2n)}.$

Remark on the rigor of the approximation. A more rigorous way to justify the approximation is to take logarithms:

$$\ln \Pr[\text{no collision}] = \sum_{j=1}^{m-1} \ln \left(1 - \frac{j}{n} \right).$$

Using

$$\ln(1-x) = -x + O(x^2),$$

we get

$$\ln \Pr[\text{no collision}] = - \sum_{j=1}^{m-1} \frac{j}{n} + O \left(\sum_{j=1}^{m-1} \frac{j^2}{n^2} \right).$$

Since

$$\sum_{j=1}^{m-1} j = \frac{m(m-1)}{2}$$

and

$$\sum_{j=1}^{m-1} j^2 = O(m^3),$$

we obtain

$$\ln \Pr[\text{no collision}] = - \frac{m(m-1)}{2n} + O \left(\frac{m^3}{n^2} \right).$$

Thus, for example when $m = O(\sqrt{n})$, the error term is $O(n^{-1/2}) = o(1)$, and hence

$$\Pr[\text{no collision}] = \exp \left(- \frac{m(m-1)}{2n} + o(1) \right) \approx e^{-m^2/(2n)}.$$